chapter-13

2) $f(x) = |x|$

when, $x>0$, $f(x) = |x|$
$= x$

$$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0+h)$$

$$= \lim_{h\to 0} f(0+h)$$

$$= \lim_{h\to 0} (0+h)$$

$$= 0$$

when, $x<0$ $f(x) = |x|$
$= x$

$$\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h)$$

$$= \lim_{h\to 0} (0-h)$$

$$= 0$$

when, $x=0$; $f(x) = |x|$

$\therefore f(0) = |0|$
$= 0$

$$\therefore \lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0)$$

$\therefore$ Thus $f(x)$ is continuous at $x=0$.

b) $f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases}$

when, $x > 0$ $f(x) = x$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} (0+h)$$

$$= 0$$

when, $x < 0$, $f(x) = -x$

$$\lim_{x \to 0^-} f(x) = \cancel{f(0-h)} \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} -(0-h)$$

$$= 0$$

when, $x = 0$, $f(x) = x$

$\therefore$ $f(0) = 0$

Thus $f(x)$ is continous at $x = 0$.

(v) $f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ \frac{x}{1} & \text{when } x = 0 \end{cases}$

Solution:-

when, $x > 0$, $f(x) = \frac{|x|}{x}$

$$\lim_{x \to 0^+} f(x) = \cancel{f(x)} \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} \cancel{\frac{|(0+h)|}{(0+h)}} = \lim_{h \to 0} f(h)$$

$$= \lim_{h \to 0} \frac{|h|}{h}$$

$$= \lim_{h \to 0} 1$$

$$= 1$$

when, $x < 0$, $f(x) = \frac{-|x|}{x} = -1$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} f(-h)$$

$$= \lim_{h \to 0} -1$$

$$= -1$$

when, $x = 0$, $f(x) = 1$

$\therefore f(0) = 1$

$$\lim_{x \to 0^+} f(x) \neq \lim_{x \to 0^-} f(x) \neq \lim_{x \to 0} f(0)$$

Thus $f(x)$ is not continous at $x = 0$.

২২। $f(x) = \begin{cases} x, & 0 \le x < \frac{1}{2} \\ 1-x, & \frac{1}{2} \le x \le 1 \end{cases}$

when, $x < \frac{1}{2}$, $f(x) = x$

$$\lim_{x \to \frac{1}{2}^-} f(x) = \lim_{h \to 0} f\left(\tfrac{1}{2} - h\right)$$

$$= \lim_{h \to 0} \left(\tfrac{1}{2} - h\right)$$

$$= \frac{1}{2}$$

when, $x > \frac{1}{2}$, $f(x) = 1 - x$

$$\lim_{x \to \frac{1}{2}^+} f(x) = \cancel{f\left(\tfrac{1}{2}+h\right)} \lim_{h \to 0} f\left(\tfrac{1}{2} + h\right)$$

$$= \lim_{h \to 0} \left(\tfrac{1}{2} + h\right)$$

$$= \tfrac{1}{2}$$

when, $\cancel{f}\ x = \frac{1}{2}$, $f(x) = 1 - x$

$\cancel{\lim_{x \to 1}}$

$$\therefore f\left(\tfrac{1}{2}\right) = 1 - \tfrac{1}{2}$$

$$= \tfrac{1}{2}$$

Thus $f(x)$ is continuous at $x = \frac{1}{2}$.

$$R.f'\left(\tfrac{1}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{1}{2} + h\right) - f\left(\frac{1}{2}\right)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{2} - h - \frac{1}{2}}{h} = \lim_{h \to 0} \frac{1 - 2h - 1}{2h}.$$

$$= \lim_{h \to 0} -1$$

$$= -1$$

$$L \cdot f'(1/2) = \lim_{h \to 0} \frac{f(1/2 - h) - f(1/2)}{-h}$$

$$= \lim_{h \to 0} \frac{1/2 - h - 1/2}{-h}$$

$$= \lim_{h \to 0} 1$$

$$= 1$$

$\therefore Rf'(1/2) \neq L \cdot f'(1/2)$

Thus $f(x)$ is not differentiable at $x = 1/2$.

28) $f(x) = \begin{cases} 1 , & -\infty < x < 0 \\ 1 + \sin x , & 0 \leq x \leq \pi/2 \\ 2 + (x - \frac{\pi}{2})^2 , & \frac{\pi}{2} \leq x \leq \infty \end{cases}$

Solution:-

At, $x = \pi/2$

when, $x > \pi/2$, $f(x) = 2 + (x - \pi/2)^2$

$$\lim_{x \to \frac{\pi}{2}^+} f(x) = \lim_{h \to 0} f(\pi/2 + h)$$

$= \lim_{h \to 0} 2 + \left(\frac{\pi}{2} + h - \pi/2\right)^2$

$= \lim_{h \to 0} (2 + h^2)$

$= 2$

when, $x < \frac{\pi}{2}$, $f(x) = 1 + \sin x$

$\lim_{x \to \frac{\pi}{2}^-} f(x) = \cancel{f(\frac{\pi}{2} - h)} \lim_{h \to 0} f(\frac{\pi}{2} - h)$

$= \lim_{h \to 0} 1 + \sin(\frac{\pi}{2} - h)$

$= \lim_{h \to 0} (1 + \cos h)$

$= 2$

when, $x = \frac{\pi}{2}$, $f(x) = 1 + \sin x$

$f(\pi/2) = 1 + \sin \pi/2$

$= 2$

$\therefore \lim_{x \to \frac{\pi}{2}^+} f(x) = \lim_{x \to \pi/2^-} f(x) = f(\pi/2)$

$\therefore$ Thus $f(x)$ is countinuos at $x = \pi/2$.

$Rf'(\frac{\pi}{2}) = \lim_{h \to 0} \frac{f(\pi/2 + h) - f(\pi/2)}{h}$

$= \lim_{h \to 0} \frac{2 + h^2 - 2}{h}$

$$= \lim_{h\to 0} (h)$$

$$= 0$$

$$L.f'(\pi/2) = \lim_{h\to 0} \frac{f(\pi/2 - h) - f(\pi/2)}{-h}$$

$$= \lim_{h\to 0} \frac{1+\cos h - 2}{-h}$$

$$= \lim_{h\to 0} \frac{1-\cos h}{h}$$

$$= \lim_{h\to 0} \frac{1-\cos 2 \cdot h/2}{h}$$

$$= \lim_{h\to 0} \frac{2\sin^2 h/2}{h}$$

$$= \lim_{h\to 0} \frac{\sin h/2}{h/2} \times \lim_{x\to 0} \sin h/2$$

$$= 1 \times 0$$

$$= 0$$

$$\therefore Rf'(\pi/2) = Lf'(\pi/2)$$

Hence, $f(x)$ is differentiable at $x = \pi/2$

At, $x = 0$

when, $x > 0$, $f(x) = 1 + \sin x$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} \{1 + \sin(0+h)\}$$

$$= \lim_{h \to 0} (1 + \sin h)$$

$$= 1$$

when, $x < 0$, $f(x) = 1$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} 1$$

$$= 1$$

when, $x = 0$, $f(x) = 1 + \sin x$

$$\therefore f(0) = 1 + \sin 0$$

$$= 1 + 0$$

$$= 1$$

$$\therefore \lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = \cancel{\lim_{x \to}} f(0)$$

$\therefore$ Thus ~~the~~ $f(x)$ is continuous at $x = 0$.

$$Rf'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0} \frac{1+\sin h - 1}{h}$$

$$= \lim_{h\to 0} \frac{\sin h}{h}$$

$$= \lim_{h\to 0} 0$$

$$Lf'(0) = \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h\to 0} \frac{1-1}{-h}$$

$$= 0$$

$\therefore Rf'(0) \neq Lf'(0)$

Hence, $f(x)$ is not differentiable at $x = 0$

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$$f(x) = \begin{cases} 3+2x, & -\frac{3}{2} \leq x \leq 0 \\ 3-2x, & 0 \leq x \leq \frac{3}{2} \\ -3-2x, & x \geq \frac{3}{2} \end{cases}$$

Solution:-

when, $x > \frac{3}{2}$, $f(x) = -3-2x$

$\therefore \lim_{x\to\frac{3}{2}^+} f(x) = \lim_{h\to 0} f\cancel{(x)}(\frac{3}{2}+h)$

$= \lim_{h\to 0} \{-3-2(\frac{3}{2}+h)\}$

$= \lim_{h\to 0} (-3-3-2h)$

$= \lim_{h\to 0} (-6-2h)$

$= -6$

when, $x < \frac{3}{2}$, $f(x) = 3-2x$

$\therefore \lim_{x\to\frac{3}{2}^-} f(x) = \lim_{h\to 0} f(\frac{3}{2}-h)$

$= \lim_{h\to 0} \{3-2(\frac{3}{2}-h)\}$

$= \lim_{h\to 0} (3-3+2h)$

$= \lim_{h\to 0} (2h)$

$= 0$

when, $x = \frac{3}{2}$, $f(x) = 3-2x$

$\therefore f(3/2) = 3 - 2\cdot 3/2 = 0$

$\therefore \lim_{x \to \frac{3}{2}^+} \neq \lim_{x \to \frac{3}{2}^-} f(x) = \lim f(3/2)$.

Thus f(x) is not continous at $x = 3/2$.

$$Rf'\left(\frac{3}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{3}{2}+h\right) - f(3/2)}{h}$$

$$= \lim_{h \to 0} \frac{-6-2h-0}{h}$$

$$= \lim_{h \to 0} \frac{-6-2h}{h}$$

$$= -\infty$$

$$L \cdot f'(3/2) \lim_{h \to 0} \frac{f\left(\frac{3}{2}-h\right) - f(3/2)}{-h}$$

$$= \lim_{h \to 0} \frac{0 \cancel{-} 0}{-h} = \lim_{h \to 0} \frac{2h - 0}{-h}$$

$$= \cancel{\lim_{h \to 0}}\ \phi \qquad = \lim_{h \to 0} \frac{2h}{-h}$$

$$= \lim_{h \to 0} (-2)$$

$$= -2$$

$\therefore Rf'(3/2) \neq Lf'(3/2)$

Hence ~~the~~ $f'(x)$ is not differentiable at $x = 3/2$.

69) $f(x) = \dfrac{x^2+3x-10}{x-2}, \quad x \neq 2$

$\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h)$

$= \lim_{h\to 0} \dfrac{(2+h)^2+3(2+h)-10}{(2+h)-2}$

$= \lim_{h\to 0} \dfrac{4+4h+h^2+6+3h-10}{h}$

$= \lim_{h\to 0} \dfrac{h^2+7h}{h}$

$= \lim_{h\to 0} \dfrac{h(h+7)}{h}$

$= \lim_{h\to 0} (h+7)$

$= 7.$

$\lim_{x\to 2^-} f(x) = \lim_{h\to 0} (2-h)$

$= \lim_{h\to 0} \dfrac{(2-h)^2+3.(2-h)-10}{(2-h)-2}$

$= \lim_{h\to 0} \dfrac{4-4h+h^2+6-3h-10}{2-h-2}$

$= \lim_{h\to 0} \dfrac{h^2-7h}{-h}$

$= \lim_{h\to 0} 7-h$

$= 7$

$$\therefore \lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} f(x) = 7$$

The function will be continaous $x=2$, if $f(2) = 7$.

General Theorem

1. Rolle's theorem:-

Statement:-

If, (i) $f(x)$ is continuous in the closed interval $a \leq f(x) \leq b$.

(ii) $f(x)$ is differential in a open interval $a < f(x) < b$.

(iii) $f(a) = f(b)$.

then there exist at leaste one value of x (say c) between a and b, where $f'(c) = 0$.

proof:

Since, $y = f(x)$ is continuous in a closed interval $a \leq x \leq b$. So, the function has maximum or minimum or both values in the interval $a \leq x \leq b$. let, the maximum or minimum value of $y = f(x)$ are M and m respectfuly,

Now, $M \neq m$ or $M = m$,

when,

$M \neq m$, $f(x) = \text{Constant}$

$$f'(x) = 0$$

so, there is nothing prove.

If,

$M = m$, then there exist at least one point between a and b, where $f(c) = m$, that is $f(c)$ is the minimum value at $x = c$.

$\therefore \lim_{h\to 0^+} f(c+h) \geq f(c)$

$\Rightarrow \lim_{h\to 0^+} \{f(c+h) - f(c)\} \geq 0$

$\Rightarrow \lim_{h\to 0^+} \dfrac{f(c+h) - f(c)}{h} \geq 0$

$\therefore Rf'(c) \geq 0$ —— (i)

and,

$\lim_{h\to 0^-} f(c+h) \geq f(c)$

$\Rightarrow \lim_{h\to 0^-} \{f(c+h) - f(c)\} \geq 0$

$\Rightarrow \lim_{h\to 0^-} \dfrac{\{f(c+h) - f(c)\}}{-h} \geq 0$

$\therefore Lf'(c) \geq 0$ —— (ii)

~~Again, there~~

From (i) & (ii)

$f'(c) = 0$

Again, there exist one point $x=c$ between a and b, where, $f(x)=M$, that is $f(c)$ is the maximum value at $x=c$.

$$\therefore \lim_{h\to 0^+} f(c+h) \geq f(c)$$

$$\Rightarrow \lim_{h\to 0^+} \{f(c+h)-f(c)\} \geq 0$$

$$\Rightarrow \lim_{h\to 0^+} \frac{\{f(c+h)-f(c)\}}{h} \geq 0$$

$$\therefore R.f'(c) \geq 0 \text{ ——— (iii)}$$

and,

$$\lim_{h\to 0^-} f(c+h) \geq f(c)$$

$$\Rightarrow \lim_{h\to 0^-} \{f(c+h)-f(c)\} \geq 0$$

$$\Rightarrow \lim_{h\to 0^-} \frac{\{f(c+h)-f(c)\}}{-h} \geq 0$$

$$\therefore L.f'(c) \geq 0 \text{ ——— (iv)}$$

From, (iii), (iv),

$$f'(c)=0$$

(proved)

2. Cauchy's mean value theorem:

Statement:-

If two functions $f(x)$ and $g(x)$ statisfying the following Conditions:-

1. $f(x)$ and $g(x)$ are continuous in a closed interval $a \leq x \leq b$.

2. $f(x)$ and $g(x)$ are differential in a open interval $a < x < b$.

3. $g'(x) \neq 0$ and $g(a) = g(b)$.

then, there exist at least one ~~point~~ ~~(say $x = c$)~~ value of x (say $x = c$) between a and b, i.e $a < c < b$, then.

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

proof:-

Considering, $\phi(x) = f(x) - Ag(x)$ —— (i)

Where, $\phi(x)$ satisfy all the properties of Rolle's theorem and A is any constant.

$$\therefore \varphi(a) = \varphi(b)$$

$$\Rightarrow f(a) - Ag(a) = f(b) - Ag(b)$$

$$\Rightarrow Ag(b) - Ag(a) = f(b) - f(a)$$

$$\Rightarrow A\{g(b) - g(a)\} = f(b) - f(a)$$

$$\therefore A = \frac{f(b) - f(a)}{g(b) - g(a)} \text{ ——— (ii)}$$

Since, $\varphi(x)$ satisfy Rolle's theorem, so, $\varphi'(c) = 0$.

Now, from (i),

$$\varphi'(c) = f'(c) - A \cdot g'(c)$$

$$\Rightarrow 0 = f'(c) - Ag'(c)$$

$$\therefore A = \frac{f'(c)}{g'(c)} \text{ ——— (iii)}$$

From (ii) and (iii) we get,

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

(proved)

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3. (b) Here,

$f(x) = x^2$

$\therefore f'(x) = 2x$

$\therefore f(-1) = (-1)^2 = 1$

$f(1) = 1^2 = 1$

$\therefore f(-1) = f(1)$

Again, $R.f'(x) = \lim_{h \to 0^+} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0^+} \frac{(x+h)^2 - x^2}{h}$

$= \lim_{h \to 0^+} \frac{x^2 + 2xh + h^2 - x^2}{h}$

$= \lim_{h \to 0^+} \frac{h(2x+h)}{h}$

$= \lim_{h \to 0^+} (2x+h)$

$= 2x.$

$L.f'(x) = \cancel{f(x} \lim_{h \to 0^-} \frac{f(x-h) - f(x)}{-h}$

$= \lim_{h \to 0} \frac{(x-h)^2 - x^2}{-h}$

$= \lim_{h \to 0} \frac{x^2 - 2xh - h^2 - x^2}{-h}$

$= \lim_{h \to 0} \dfrac{-h(2x+h)}{-h}$

$= \lim_{h \to 0} (2x+h)$

$= 2x$

$\therefore Rf'(x) = Lf'(x)$. So, $f(x)$ is differential between $(-1,1)$.
then. $f(x)$ is continuous within $[-1,1]$,

According to Rolle's theorem,

$f'(c) = 0$

$\Rightarrow 2c = 0$

$\therefore c = 0 \in (-1,1)$

(proved)

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3. (b) Here,

$f(x) = 3+2x-x^2$

$\therefore f'(x) = \cancel{3}\ 2-2x$

$\therefore f(0) = 3.$

$f(1) = 3+2-1 = 3+1 = 4$

$$R.f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{3+2(x+h)-(x+h)^2-(3+2x-x^2)}{h}$$

$$= \lim_{h\to 0} \frac{3+2x+2h-x^2-2hx-h^2-3-2x+x^2}{h}$$

$$= \lim_{h\to 0} \frac{2h-2hx-h^2}{h}$$

$$= \lim_{h\to 0} \frac{h(2-2x-h^2)}{h}$$

$$= \lim_{h\to 0} (2-2x-h^2)$$

$$= 2-2x$$

$$L.f'(x) = \lim_{h\to 0} \frac{f(x-h)-f(x)}{-h}$$

$$= \lim_{h\to 0} \frac{3+2(x-h)-(x-h)^2-(3+2x-x^2)}{-h}$$

$$= \lim_{h\to 0} \frac{3+2x-2h-(x^2-2xh+h^2)-3-2x+x^2}{-h}$$

$$= \lim_{h\to 0} \frac{3+2x-2h-x^2+2xh-h^2-3-2x+x^2}{-h}$$

$$= \lim_{h\to 0} \frac{-2h+2xh-h^2}{-h}$$

$$= \lim_{h\to 0} \frac{-h(2-2x+h)}{-h}$$

$= \lim_{h\to 0} (2-2x+h)$

$= 2-2x$

$\therefore$ R.$f'(x) =$ L.$f'(x)$. So

Now,

$f'(c) = 2-2c$

And,

$\dfrac{f(1)-f(0)}{1-0}$

$= \dfrac{4-3}{1}$

$= 1/1 = 1.$

Here,

$f'(c) = 1$

$\Rightarrow 2-2c = 1$

$\Rightarrow -2c = 1-2$

$\Rightarrow c = \dfrac{-1}{-2}$

$= 1/2$

$\therefore 0 < 1/2 < 1.$

$\therefore f'(c) = 2-2c = 2-2\cdot 1/2 = 2-1 = 1 = \dfrac{f(1)-f(0)}{1-0}$

(proved)